

REPORT

Title

Prolog Program to Sum Integers from 1 to N

Aim

To write and execute a Prolog program to find the sum of integers from **1** to **N** using recursion.

Problem Statement

Write a Prolog program that calculates the sum of all integers from 1 to a given positive integer N.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (exp1.pl)

Procedure

1. Open **SWI-Prolog**.
2. Create a new Prolog file named exp1.pl.
3. Enter the recursive sum program.
4. Save the file.
5. Load the program using [exp1] ..
6. Execute the query sum(10, S) ..
7. Observe the calculated sum.

Program

```
sum(0, 0) .
```

```
sum(N, S) :-
```

```
N > 0,  
N1 is N - 1,  
sum(N1, S1),  
S is S1 + N.
```

Sample Output

Query:

```
?- [exp1].  
true.
```

```
?- sum(10, S).  
S = 55.
```

Another example:

```
?- sum(5, S).  
S = 15.
```

Conclusion

Thus, the Prolog program was successfully implemented using recursion to calculate the **sum of integers from 1 to N**. For $N = 10$, the result obtained is **55**.